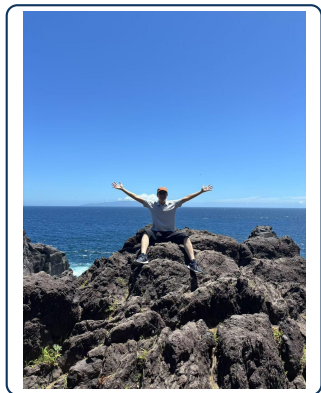


Stationary MMD Points

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Research Interests

- Kernel and nonparametric methods
- Statistical learning theory

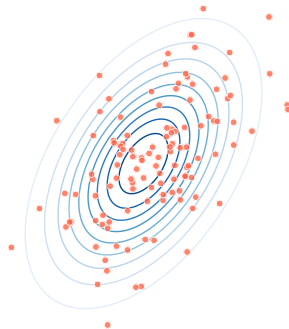
Motivation

Problem: Approximation of a target distribution μ with a finite set of points $\{x_i\}_{i=1}^n$.

- Denote the empirical distribution of $\{x_i\}_{i=1}^n$ as

$$\mu_n = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}.$$

- We want μ_n to be close to μ .

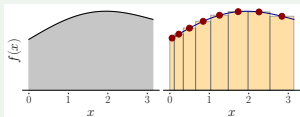


Motivation

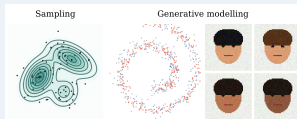
Problem: Approximation of a target distribution μ with a finite set of points $\{x_i\}_{i=1}^n$.



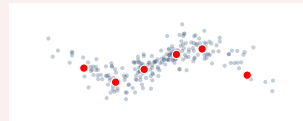
Numerical Integration



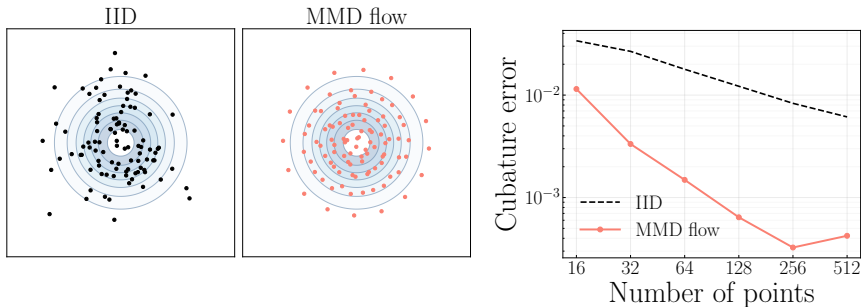
Sampling



Coreset Selection



An empirical observation



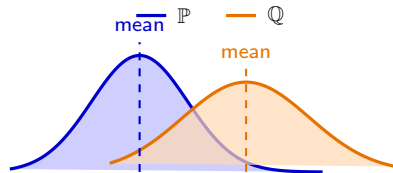
- Samples from MMD flow are more **uniform** than IID samples [XKS22].
- Lower numerical integration error for fixed n .

$$\left| \int f(x) \, d\mu(x) - \int f(x) \, d\mu_n(x) \right|.$$

Goal: Why?

Background: What is MMD?

- Given two distributions $\mathbb{P}$ and $\mathbb{Q}$, how to compare them?
- $|\mathbb{E}_{\mathbb{P}}[X] - \mathbb{E}_{\mathbb{Q}}[X]|$.
- $|\mathbb{E}_{\mathbb{P}}[X^2] - \mathbb{E}_{\mathbb{Q}}[X^2]|$.
- The first two moments are not enough!



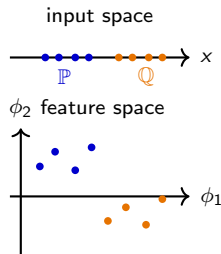
Goal: Compare $\mathbb{P}$ and $\mathbb{Q}$ with a general f , i.e., $|\mathbb{E}_{\mathbb{P}}[f(X)] - \mathbb{E}_{\mathbb{Q}}[f(X)]|$?

Background: Maximum Mean Discrepancy (MMD) [GBR⁺12]

- $\mathcal{H}$ is a reproducing kernel Hilbert space (RKHS) with kernel k .

$$\text{MMD}(\mathbb{P}, \mathbb{Q}) := \sup_{f \in \mathcal{H}} |\mathbb{E}_{\mathbb{P}}[f(X)] - \mathbb{E}_{\mathbb{Q}}[f(X)]|.$$

- The RKHS contains functions that are linear combinations of the feature maps $\phi : \mathcal{X} \rightarrow \mathcal{H}$, $f = \sum_{i=1}^n \alpha_i \phi(x_i)$.



Theorem

MMD admits a closed-form expression:

$$\text{MMD}^2(\mathbb{P}, \mathbb{Q}) = \mathbb{E}_{X, X' \sim \mathbb{P}}[k(X, X')] + \mathbb{E}_{Y, Y' \sim \mathbb{Q}}[k(Y, Y')] - 2\mathbb{E}_{X \sim \mathbb{P}, Y \sim \mathbb{Q}}[k(X, Y)].$$

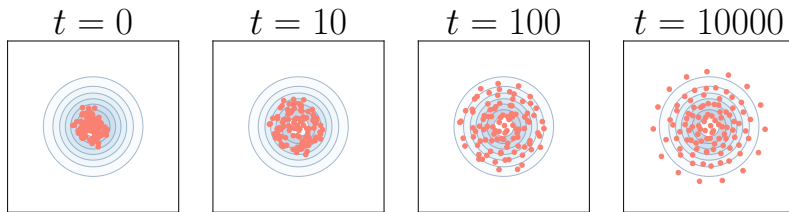
- MMD is a probability divergence ($\text{MMD}(\mathbb{P}, \mathbb{Q}) = 0 \iff \mathbb{P} = \mathbb{Q}$) if k is c_0 -universal (Gaussian) [SFL11].

Background: MMD gradient flow [AKSG19]

- Start with an initial set of n particles $\{x_i^{(0)}\}_{i=1}^n$ at time 0.
- Evolve the particles according to the MMD gradient flow:

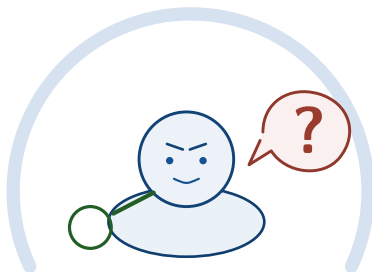
$$x_i^{(t+1)} = x_i^{(t)} - \gamma \cdot \nabla_{x_i^{(t)}} \text{MMD}^2 \left(\mu, \frac{1}{n} \sum_{j=1}^n \delta_{x_j^{(t)}} \right), \quad i = 1, \dots, n.$$

- $\gamma > 0$ is the step size.



- A Wasserstein gradient flow.

Observation: MMD flow samples are more **uniform** than IID samples [XKS22].



Existing work

- Existing finite-sample rates in MMD gradient flows [AKSG19, CMG⁺25].
- Requires noise injection.
- Let $\mu_n^{(t)} = \frac{1}{n} \sum_{j=1}^n \delta_{x_j^{(t)}}$ be the empirical distribution of n particles at iterate t .

$$\text{MMD} \left(\mu, \mu_n^{(t)} \right) \leq C(t) n^{-\frac{1}{2}}, \quad W_2 \left(\mu, \mu_n^{(t)} \right) \leq C(t) n^{-\frac{1}{2}}.$$

Three limitations

- Same rate as IID sampling: $n^{-\frac{1}{2}}$.
- The constant $C(t)$ grows exponentially in t .
- Noise injection.

Existing work

- Minimum MMD points

$$\{x_i^*\}_{i=1}^n = \arg \min_{\{x_i\}_{i=1}^n} \text{MMD}^2 \left(\mu, \frac{1}{n} \sum_{j=1}^n \delta_{x_j} \right).$$

- Let $\mu_n^* = \frac{1}{n} \sum_{j=1}^n \delta_{x_j^*}$ be the empirical distribution [XKS22, DP10]:

$$\text{MMD}(\mu, \mu_n^*) \leq Cn^{-1}.$$

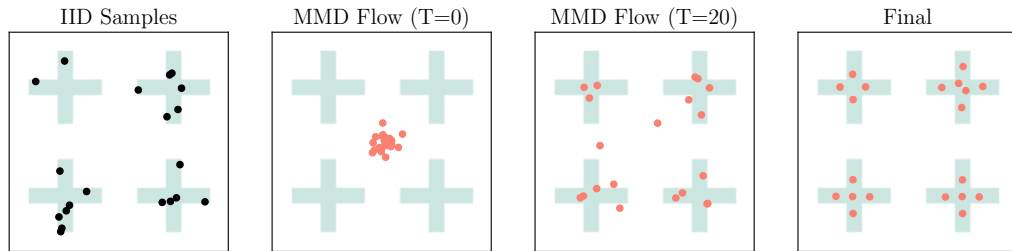
- Quasi Monte Carlo point set achieve this rate [DP10].

Two limitations

- Minimum MMD points are hard to compute, due to the non-convexity of MMD.
- Gradient descent only finds a **stationary** point instead.

Motivation

✗ ~~Minimum MMD points~~ but ✓ **Stationary MMD points**



- A four-cross distribution.
- An ideal minimum MMD points would have 5 points in each cross.
- Stationary MMD points have (4, 6, 5, 5) points in each cross.

Stationary MMD points

Definition (Stationary MMD points)

A sequence of point sets $\{x_i\}_{i=1}^n$ with $\mu_n := \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$, that satisfies:

1. **Convergence:** $\text{MMD}(\mu, \mu_n) \rightarrow 0$ as $n \rightarrow \infty$.
2. **Stationarity:** $\nabla_{x_1, \dots, x_n} \text{MMD}^2(\mu, \mu_n) = 0$.

- Minimum MMD points are a special case of stationary MMD points.
- Stationary MMD points are easier to compute, e.g., via MMD gradient flow.

Question: Why does stationary MMD points exhibit superior numerical integration performance than IID points?

Stationary MMD points

- Simulate MMD flow for long enough, $t \rightarrow \infty$.
- For any $i \in \{1, \dots, n\}$:

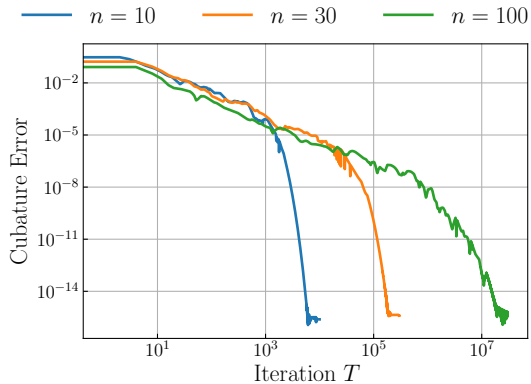
$$\nabla_{x_i} \text{MMD}^2 \left(\mu, \frac{1}{n} \sum_{j=1}^n \delta_{x_j} \right) = 0 \implies \frac{1}{n} \sum_{j=1}^n \nabla_1 k(x_i, x_j) - \int \nabla_1 k(x_i, y) d\mu(y) = 0.$$

- Exact integration on $x \mapsto \nabla_1 k(x_i, x)$.
- Exact integration on $\{1\}$.

Exact integration subspace:

$$\mathcal{F}_n := \text{span} \left(\{1\} \cup \{ \nabla_1 k(x_i, \cdot) : i \in \{1, \dots, n\} \} \right) \subset \mathcal{H}.$$

Stationary MMD points



- Exact integration on $\mathcal{F}_n$.

Stationary MMD points

Assumption

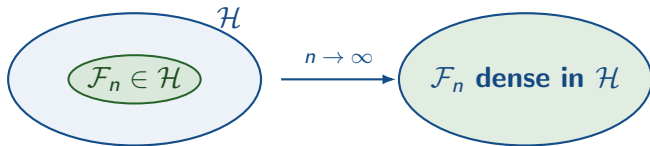
The kernel k is smooth and c_0 -universal.

Proposition (Approximation of $\mathcal{F}_n$ towards $\mathcal{H}$)

Given $f \in \mathcal{H}$. Consider the semi-norm

$$|f|_n := \inf \{ \|r_n\|_{\mathcal{H}} : f = f_n + r_n, f_n \in \mathcal{F}_n, r_n \in \mathcal{H} \}$$

Then, $\lim_{n \rightarrow \infty} |f|_n = 0$.



Stationary MMD points

Theorem (Super-convergence)

For a stationary MMD point set $\{x_i\}_{i=1}^n$ and $\mu_n = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$,

$$\left| \frac{1}{n} \sum_{i=1}^n f(x_i) - \int f \, d\mu \right| \leq |f|_n \cdot \text{MMD}(\mu, \mu_n)$$

Proof sketch

- Decompose $f = f_n + r_n$ with $f_n \in \mathcal{F}_n$.

$$\left| \frac{1}{n} \sum_{i=1}^n f(x_i) - \int f \, d\mu \right| = \left| \frac{1}{n} \sum_{i=1}^n r_n(x_i) - \int r_n \, d\mu \right| \leq \|r_n\|_{\mathcal{H}} \cdot \text{MMD}(\mu, \mu_n),$$

Stationary MMD points

Theorem (Super-convergence)

Let $f \in \mathcal{H}$ be fixed. For a stationary MMD point set $\{x_i\}_{i=1}^n$ and $\mu_n = \frac{1}{n} \sum_{i=1}^n \delta_{x_i}$,

$$\left| \frac{1}{n} \sum_{i=1}^n f(x_i) - \int f \, d\mu \right| \leq |f|_n \cdot \text{MMD}(\mu, \mu_n).$$

$\rightarrow 0$

Stationarity

$\rightarrow 0$

MMD flow

- Proved already (✓): $|f|_n \rightarrow 0$ due to stationarity.
- Next step (→): $\text{MMD}(\mu, \mu_n) \rightarrow 0$.
- Key bit (★): Avoid $C(t)$ exponential in time.

If we can prove that, $\text{MMD}(\mu, \mu_n) = \mathcal{O}(n^{-\frac{1}{2}})$, then we obtain $o(n^{-\frac{1}{2}})$ rate.

Stationary MMD points

- At iterate $t \in \mathbb{N}$
- MMD gradient flow: for $i = 1, \dots, n$,

$$x_i^{(t+1)} = x_i^{(t)} - \gamma \left\{ \frac{1}{n} \sum_{j=1}^n \nabla_1 k(x_i^{(t)}, x_j^{(t)}) - \mathbb{E}_{Y \sim \mu} [\nabla_1 k(x_i^{(t)}, Y)] \right\}$$

- **Noisy** MMD gradient flow: for $i = 1, \dots, n$,
 - $u_i^{(t)} \sim \mathcal{N}(0, I_d)$ and $\beta_t > 0$.

$$x_i^{(t+1)} = x_i^{(t)} - \gamma \left\{ \frac{1}{n} \sum_{j=1}^n \nabla_1 k(x_i^{(t)} + \beta_t u_i^{(t)}, x_j^{(t)}) - \mathbb{E}_{Y \sim \mu} [\nabla_1 k(x_i^{(t)} + \beta_t u_i^{(t)}, Y)] \right\}$$

Stationary MMD points

Assumption (Noise level)

Define $\Phi(z, w) = \mathbb{E}_{Y \sim \mu} [\nabla_1 k(z, Y)] - \nabla_1 k(z, w)$. For each $t \in \mathbb{N}$, $\beta_t \in (0, 1)$ satisfies,

$$\frac{1}{n} \sum_{i=1}^n \mathbb{E}_{u_i^{(t)}} \left[\left\| \frac{1}{n} \sum_{j=1}^n \Phi \left(x_i^{(t)} + \beta_t u_i^{(t)}, x_j^{(t)} \right) \right\|^2 \right] \geq \beta_t^2 \text{MMD}^2 \left(\frac{1}{n} \sum_{i=1}^n \delta_{x_i^{(t)}}, \mu \right).$$

- In expectation, a gradient dominance condition.
- Seems unavoidable for convergence of MMD flow [AKSG19, CMG⁺25].



Theorem

Suppose that there exists $\beta_t \propto t^{-\alpha}$ for $0 \leq \alpha < 1/2$ such that the condition holds. C' depends only on kernel k and dimension d . Then, for any $T \leq n^{\frac{1}{\alpha}}$, with high probability,

$$\text{MMD} \left(\mu_n^{(T)}, \mu \right) = \mathcal{O} \left(\exp \left(-C' \gamma T^{1-2\alpha} \right) + \frac{(\log T)^2 T^\alpha}{\sqrt{n}} \right).$$

- Reduce the exponential dependence on T to a **polynomial** dependence.
- First end-to-end finite-particle convergence bound for MMD flow.
- Take $T \propto (\log n)^{\frac{1}{1-2\alpha}}$, then $\text{MMD} \left(\mu_n^{(T)}, \mu \right) = \mathcal{O}(n^{-\frac{1}{2}})$ up to logarithmic factors.

Stationary MMD points

$$\left| \frac{1}{n} \sum_{i=1}^n f(x_i) - \int f \, d\mu \right| \leq |f|_n \cdot \text{MMD}(\mu, \mu_n).$$

- $\lim_{n \rightarrow \infty} |f|_n = 0$ due to **stationarity**.
- $\text{MMD}(\mu, \mu_n) = \mathcal{O}(n^{-\frac{1}{2}}(\log n)^b)$ due to **MMD flow convergence**.

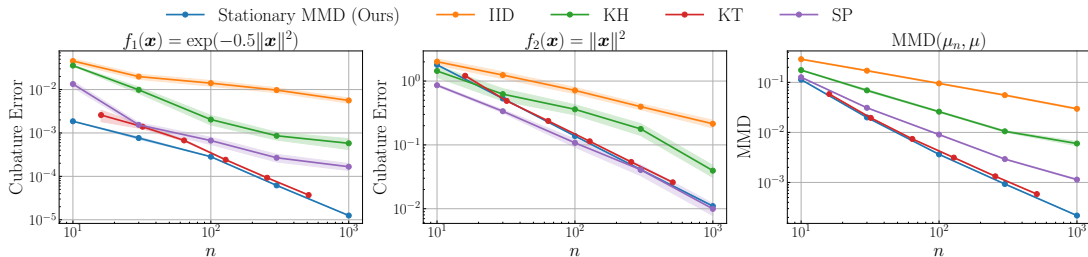


$o(n^{-\frac{1}{2}}(\log n)^b)$ rate for stationary MMD points.

Remark. If started with some point sets with $\text{MMD}(\mu, \mu_n) = \mathcal{O}(n^{-r})$, then we can run MMD flow for $T = \mathcal{O}(1)$ iterations to obtain a new point set with $o(n^{-r})$ rate.

Experiments

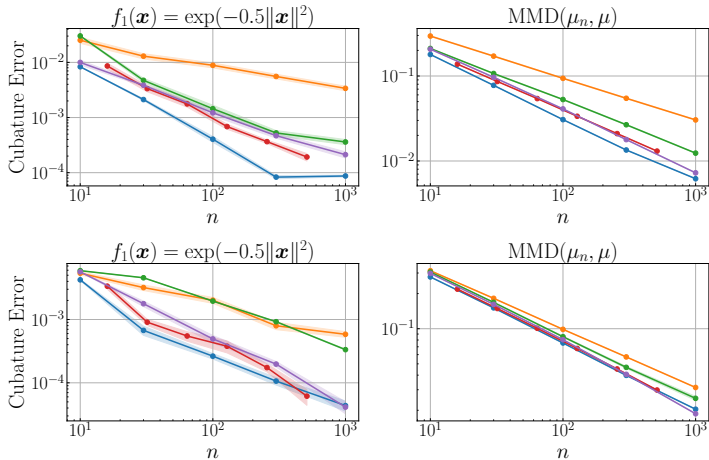
- Mixture of Gaussians.
- KH is kernel herding [CWS12]; KT is kernel thinning [DM24]; SP is support points [MJ18].



- $f_1 \in \mathcal{H}$ while $f_2 \notin \mathcal{H}$.
- Stationary MMD points along with KT are the best.
- SP works well on f_2 since it uses $k(x, y) = \|x - y\|$.

Experiments

- Coreset selection from a large dataset
- House (Top) has 22,784 and Elevator (Bottom) has 16,599 data points.



Contributions

- We propose stationary MMD points.
- A novel framework for low-discrepancy points.
- Better rate from stationarity.



Limitations

- We only prove a $o(n^{-\frac{1}{2}}(\log n)^b)$ rate, unclear if better than IID $\mathcal{O}(n^{-\frac{1}{2}})$ or not.
- Noise injection in MMD flow is not ideal.



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